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Small Project of Introduction on Machine Learning

UNDERSTANDING CUSTOMER CHARACTERISTICS AND PREDICTING CHURN IN TELECOMMUNICATIONS USING CATBOOST

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Background

Why Churn Must Be Predicted?

- In our dataset, **1 of 4** customers left the company (1,869 of 7,043 customers).
- Getting new customers costs more than keeping current customers.
- With accurate prediction, the company can act before customers leave and avoid profit loss.

Source: Wagh et al. (2024), https://doi.org/10.1016/j.rico.2023.100342

Why We Use CatBoost

- Our data is mostly categories (16 of 19 features), such as contract type and payment method.
- CatBoost is designed to handle categorical data efficiently without heavy preprocessing.
- Each new tree reduces the error of the previous trees, so predictions become more accurate and stable.
- Imbalanced data (26.5% churn) is handled with threshold tuning.

Source: Asif et al. (2025), https://doi.org/10.1016/j.rineng.2025.104629

Challenge & Idea

Challenge

Manual hyperparameter tuning is inefficient and makes it difficult to find the optimal parameter combination.

Idea

Use Optuna to optimize hyperparameters

Challenge

The dataset contains many categorical features.

Idea

Use CatBoost's native categorical feature handling.

Challenge

Raw features may not fully capture customer churn patterns.

Idea

Create relevant features through feature engineering.

Research Goals

1. To Analyze Customer Characteristics and Identify Key Drivers of Churn
2. To Develop and Optimize Machine Learning Predictive Models
3. To Evaluate Model Performance and Compare It with a Baseline Model

Proposed Method

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DATASET

IBM Telco Churn
7043 rows x 21 columns

PREPROCESSING

CLEANING
- Missing value
- Duplicate

Exploratory Data Analysis
Stayed & Churn Data Quality

Features & Encoding
44 features & 19 categoricals

DATA SPLIT

- Development — 5634 rows — 80%
- Test — 1409 rows — 20%

TUNING, VALIDATION, THRESHOLD

OPTUNA
- Search Space
- Trials & CV
- Objective Score

CV Validation
- 5-fold Validation
- OOF Prediction
- Overfitting check

Threshold OOF
- Search Threshold
- Handle Imbalance

MODEL
- Catboost ensemble
- Logistic Regression Baseline

EVALUATION
- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC-AUC

RESULT
- Top feature
- Risk Segmentation
- Learning Curve

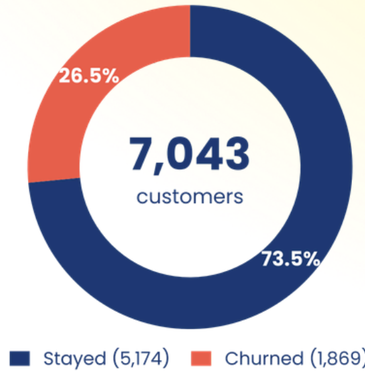
EDA (Exploratory Data Analysis)

DATASET

IBM Telco Customer Churn
7,043 rows × 21 columns

DATA QUALITY

0 duplicate rows
11 blank TotalCharges (new customers, tenure = 0)



STAYED

5,174 customers (73.5%)

CHURNED

1,869 customers (26.5%)

Numerical profile

ASPECT	STAYED	CHURNED	DIFFERENCE
Tenure (mean)	37.6 months	18.0 months	-52%, churners leave early
Monthly charges (mean)	$61.27	$74.44	+22%, churners pay more per month
Total charges (mean)	$2,549.91	$1,531.80	-40%, less accumulated spending

EDA (Exploratory Data Analysis)

Highest-risk segments

ASPECT	CATEGORY	CHURN RATE	LIFT VS AVERAGE
Tenure group	0–6 months	52.9%	2.0×
Payment method	Electronic check	45.3%	1.7×
Contract	Month-to-month	42.7%	1.6×
Internet service	Fiber optic	41.9%	1.6×
Tech support	No	41.6%	1.6×
Lowest: Contract	Two year	2.8%	0.1×

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Experimental Settings

CLEANING

- **TotalCharges** converted from text to number. 11 blank values (all tenure = 0, never billed) filled with 0.
- **Recoding:** SeniorCitizen 0/1 → No/Yes; Churn Yes/No → 1/0.
- **CustomerID** removed (unique ID, not a feature).
- No other missing values and no duplicate customer IDs.

FEATURE ENGINEERING

19 → 44 **25 new features**, created before the split from input columns only (no target used).

- 6 **Service counts** number of add-on services
- 7 **Billing behavior** average monthly charge
- 4 **Customer lifecycle** new-customer flag
- 6 **Risk profile from EDA** risk flags
- 2 **Categorical interactions** contract × payment

ENCODING

CatBoost (main model)
44 features: **19 categorical** + 25 numeric.
Categorical columns are used as they are: no one-hot, no scaling.

Logistic Regression (baseline)
One-hot encoding + standard scaling (94 columns), fitted on training folds only inside a pipeline.

DATA SPLIT

Development set (80%)
5,634 rows · 1,495 churn (26.5%)

Test set (20%)
1,409 rows · 374 churn (26.5%)
*Final evaluation only

CLASS IMBALANCE

No oversampling or class weights, so predicted probabilities are not distorted. Imbalance is handled with **PR-AUC** and **threshold tuning.**

Experimental Settings

Tuning, Validation, Threshold

OPTUNA

- TPE sampler
- **Search space:** learning_rate, depth, l2_leaf_reg, random_strength, bagging_temperature
- Score = CV AUC - Overfitting Penallty (train - val GAP)

CV Validation

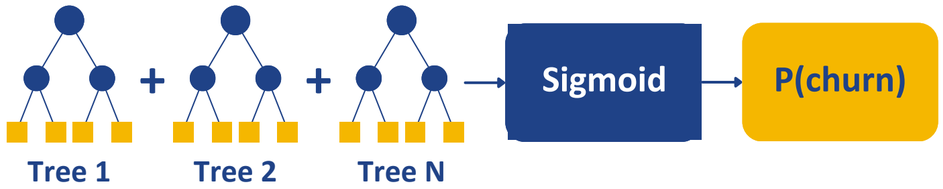
- Stratified 5-fold on dev set (final params)
- Early stopping per fold
- Overfitting check: train vs validation AUC gap
- Learning curves (iterations & data size) on dev set
- Test set stays locked

Threshold OOF

- Out-of-fold predictions from CV, not from test
- Grid 0.05 - 0.79: F1-optimal

Model

CatBoost



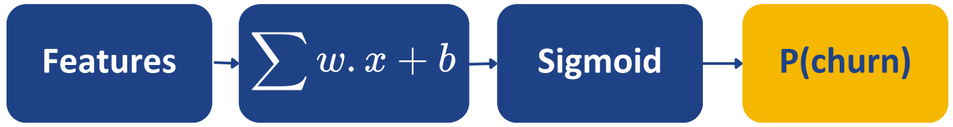
- Many trees, built one by one
- Each tree fixes earlier errors
- Non-linear, native categories

$$F(\mathbf{x}) = \sum_{t=1}^{N} \eta\, h_t(\mathbf{x})$$

$$P(\text{churn}) = \sigma\big(F(\mathbf{x})\big) = \frac{1}{1 + e^{-F}}$$

η = learning rate, ht = tree t

Logistic Regression



- One linear equation
- Weights = feature effects
- Linear, needs one-hot + scaling

$$z = \mathbf{w}^\top \mathbf{x} + b = \sum_{j=1}^{d} w_j x_j + b$$

$$P(\text{churn}) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

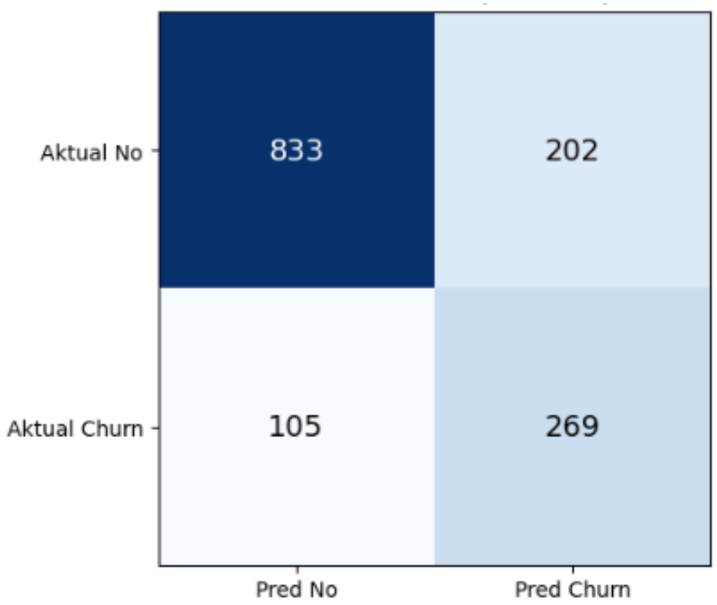
w = weights, b = intercept

Comparison

Model	Accuracy	Precision	Recall	F1 Score
Logistic regression	0.7757	0.5607	0.7166	0.6291
CatBoost	**0.7821**	**0.5711**	**0.7193**	**0.6367**

Evaluation

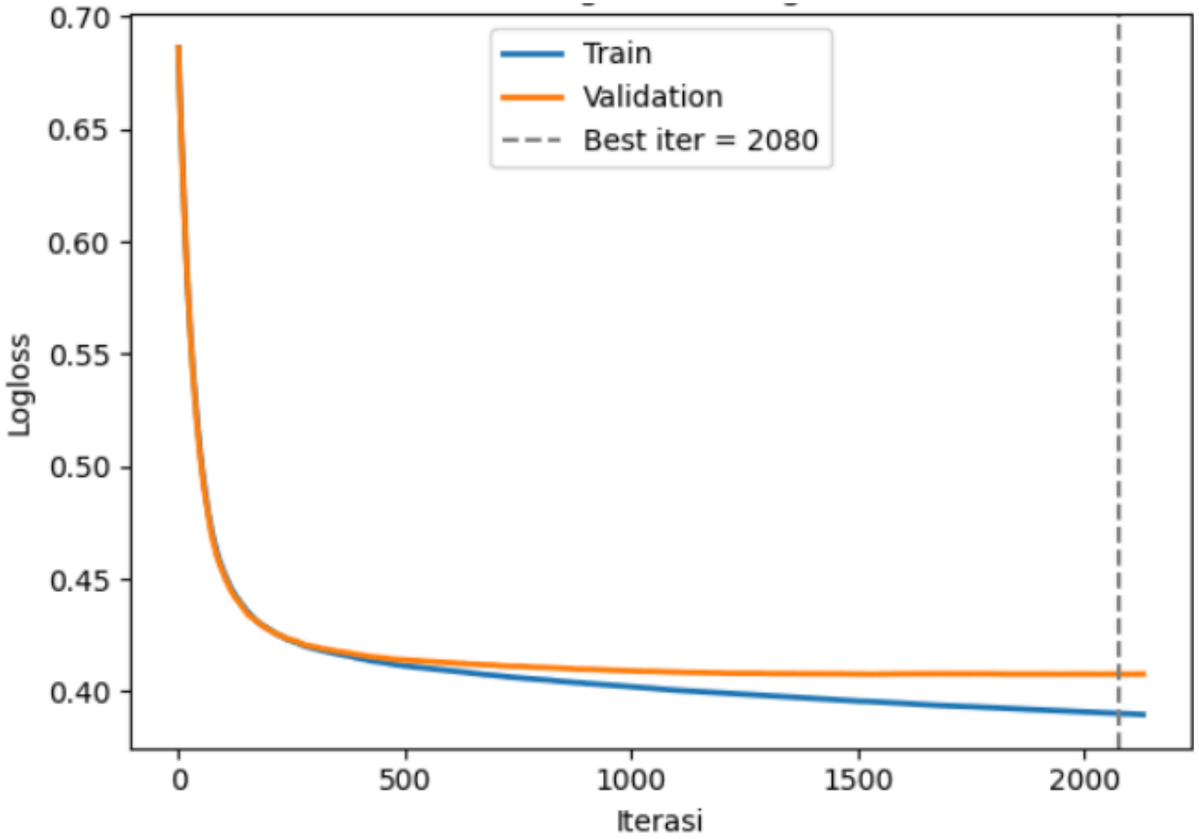
Category	Precision	Recall	F1 Score	Support
No Churn	0.888	0.805	0.844	1035
Churn	0.571	0.719	0.637	374



- The model detects **72% of churners** (recall 0.719): 269 of 374 customers.
- When it predicts churn, it is correct **57% of the time** (precision 0.571).
- Non-churn customers are predicted well (precision 0.888, recall 0.805).
- **Overall accuracy is 78.2%**, but recall and F1 for the churn class are more informative because the data is imbalanced (26.5% churn).

- **Missed churners (105) :** their profile looks like loyal customer but the data lacks behavioral signals
- **False alarms (202) :** They match high risk profile but choose to stay.

Overfitting Check



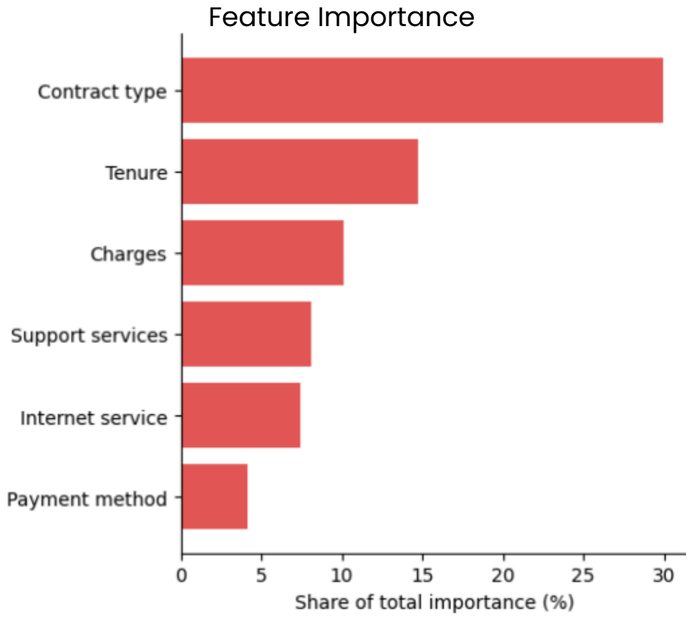
CV 5-Fold

Metric	Train	Validation	Gap
ROC-AUC	0.858	0.847	0.011
Accuracy	0.795	0.782	0.013
Logloss	0.402	0.414	0.012

- The train–validation gap is very small on all metrics (AUC gap ≈ 0.01, accuracy gap ≈ 0.013).
- Validation AUC (0.847) stays close to train AUC (0.858)

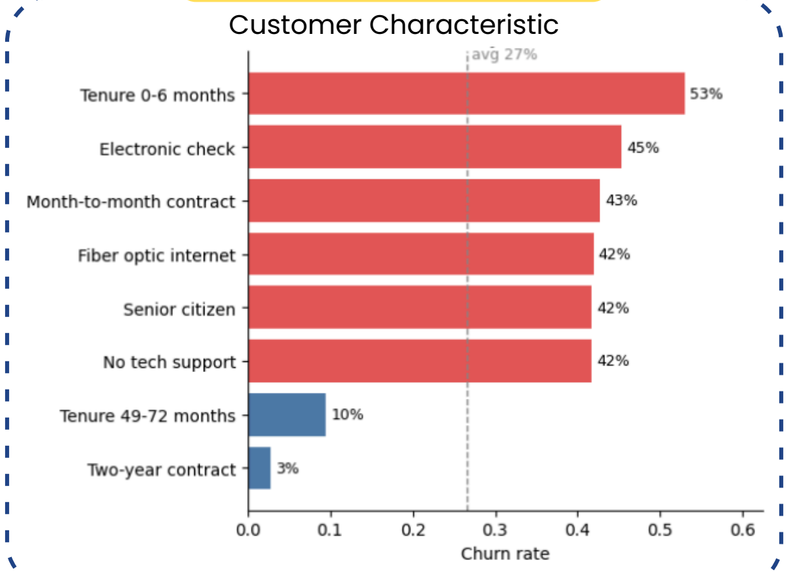
Discussion

Main Churn Drivers



- **Contract type** and **tenure** are the dominant drivers; charges, support services, internet service, and payment method contribute less

Customer Profile



- **High-risk**: new customers (0–6 months, 53%), electronic check (45%), month-to-month contracts (43%), fiber optic (42%), no tech support (42%), senior citizens (42%).
- **Low-risk:** two-year contracts (3%) and long-tenure customers (10%).

Risk Segmentation

Risk Segment	n	Average Probability	Actual Churn Rate
Low	883	0.101	0.100
Medium	321	0.444	0.445
High	205	0.711	0.698

- Actual churn rises from Low (10%) to High (70%), so the segments separate customers well.

Churn concentrates among **new customers** on flexible **month-to-month** contracts with **weak service** engagement.

Conclusion

This project analyzed customer characteristics and predicted churn using a tuned CatBoost model. Churn was highest among new customers on month-to-month contracts, and contract type and tenure were the strongest drivers. The model detected 72% of churners with 57% precision. Its predicted risk segments, from about 10% (Low) to about 70% (High) actual churn.

Recommendation

1. Encourage month-to-month customers to 1-2 years contract
2. Develop automatic payment and evaluate customer electronic check service
3. Run onboarding and programs for new customers in their first 6 months
4. Prioritize the high-risk segment in retention campaigns

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THANK U